BLUETOOTH CONTROLLED RC CAR

Technical Documentation, Wiring Diagram & Arduino Firmware

Project Name:	Smartphone RC Car	**Microcontroller:**	Arduino UNO / Nano
Document Ref:	DOC-BT-RCCAR-01	**Connectivity:**	HC-05 / HC-06 Bluetooth Module

1. PROJECT OVERVIEW & SYSTEM SPECIFICATIONS

This document serves as complete technical documentation for a **Bluetooth-Controlled RC Robotic Car**. The system uses an Arduino UNO microcontroller connected to an HC-05 (or HC-06) wireless Bluetooth module to receive directional commands from a custom smartphone application (such as *Arduino Bluetooth Controller*). Power distribution drives two or four DC gear motors via an L298N Dual H-Bridge Motor Driver module.

PARAMETER	SPECIFICATION	REMARKS
Main Controller	Arduino UNO R3 / Nano	ATmega328P Microcontroller
Motor Driver	L298N Dual H-Bridge Module	Controls up to 2 DC motors / 2 channels
Wireless Protocol	Bluetooth v2.0 + EDR (HC-05 / HC-06)	Baud Rate: 9600 bps
Operating Input Voltage	7.4V – 12V DC	2x 18650 Li-ion Batteries (7.4V) / 9V Battery
Control Range	10 Meters (Unobstructed)	Standard Bluetooth Class 2

2. COMPLETE PIN CONNECTIONS & WIRING TABLE

The following wiring matrix details all interconnections between the Arduino UNO, HC-05 Bluetooth module, L298N Motor Driver, and DC motors.

MODULE / COMPONENT	COMPONENT PIN	ARDUINO / TARGET PIN	FUNCTIONAL DESCRIPTION
HC-05 Bluetooth	VCC	5V Rail	Module Power Supply
	GND	GND Rail	Common Ground
	TXD	Pin 0 (RX) / Pin 2 (Soft Serial)	Transmits data to Arduino
	RXD	Pin 1 (TX) / Pin 3 (Soft Serial)	Receives data from Arduino (Use divider 1k/2k)
L298N Driver	IN1	Pin 8	Motor A Direction Control 1
	IN2	Pin 9	Motor A Direction Control 2
	IN3	Pin 10	Motor B Direction Control 1
	IN4	Pin 11	Motor B Direction Control 2

MODULE / COMPONENT	COMPONENT PIN	ARDUINO / TARGET PIN	FUNCTIONAL DESCRIPTION
	ENA (Optional)	Pin 5 (PWM) / Jumpered to 5V	Motor A Speed Control
	ENB (Optional)	Pin 6 (PWM) / Jumpered to 5V	Motor B Speed Control
	12V Terminal	Battery (+) Terminal	Main Power Rail (7.4V - 12V)
	GND Terminal	Battery (-) & Arduino GND	**CRITICAL: Common Ground Connection**
	5V Terminal	Arduino 5V / VIN	Powers Arduino via L298N onboard 7805 regulator
DC Motors	Motor A Left Side	L298N OUT1 & OUT2	Left side drive wheel(s)
	Motor B Right Side	L298N OUT3 & OUT4	Right side drive wheel(s)

Important Assembly & Programming Note

Always disconnect the **RX (Pin 0) and TX (Pin 1)** wires from the Arduino while uploading the code via USB! Keeping them connected during upload causes serial port communication conflict errors.

3. BILL OF MATERIALS (BOM)

ITEM	COMPONENT DESCRIPTION	QTY
1	Arduino UNO R3 / Arduino Nano Microcontroller	1
2	L298N Dual H-Bridge Motor Driver Module	1
3	HC-05 or HC-06 Bluetooth Transceiver Module	1
4	BO Motors (3V–12V Dual Axis Gear Motors) + Wheels	2 or 4
5	2-Wheel / 4-Wheel Robot Smart Car Chassis Frame	1
6	18650 Li-ion Batteries (3.7V x 2 = 7.4V) with Battery Holder	1 Set
7	Jumper Wires (Male-to-Male, Male-to-Female), SPST Switch	1 Set

4. ARDUINO FIRMWARE SOURCE CODE

The following Arduino C++ code receives Bluetooth commands via Serial interface ('F' for Forward, 'B' for Backward, 'L' for Left, 'R' for Right, and 'S' for Stop) and drives the L298N motor driver pins accordingly.

```
// --- Bluetooth Controlled RC Car Firmware ---
// Pin Declarations for L298N Driver
const int IN1 = 8;
const int IN2 = 9;
const int IN3 = 10;
```

```
const int IN4 = 11;

char command; // Variable to store incoming serial data from Bluetooth

void setup() {
  // Set motor control pins as outputs
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // Initialize Serial Communication at 9600 baud rate
  Serial.begin(9600);
  stopCar();
}

void loop() {
  if (Serial.available() > 0) {
    command = Serial.read();
    stopCar(); // Reset motor states before execution

    switch (command) {
      case 'F': moveForward(); break;
      case 'B': moveBackward(); break;
      case 'L': turnLeft();    break;
      case 'R': turnRight();   break;
      case 'S': stopCar();     break;
    }
  }
}

void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void moveBackward() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void turnLeft() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void turnRight() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void stopCar() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
```

```
  digitalWrite(IN4, LOW);
}
```

5. OPERATION & APP PAIRING INSTRUCTIONS

1. Download any **Arduino Bluetooth RC Car** app from the Google Play Store / App Store.
2. Turn ON the RC Car power switch. The LED on the HC-05 module will blink rapidly (pairing mode).
3. Pair your Android phone with HC-05 via Bluetooth settings (Default PIN: `1234` or `0000`).
4. Open the app, select the HC-05 device, and connect. The blinking LED on HC-05 will slow down to periodic double blinks.
5. Use the on-screen directional buttons to drive the car!